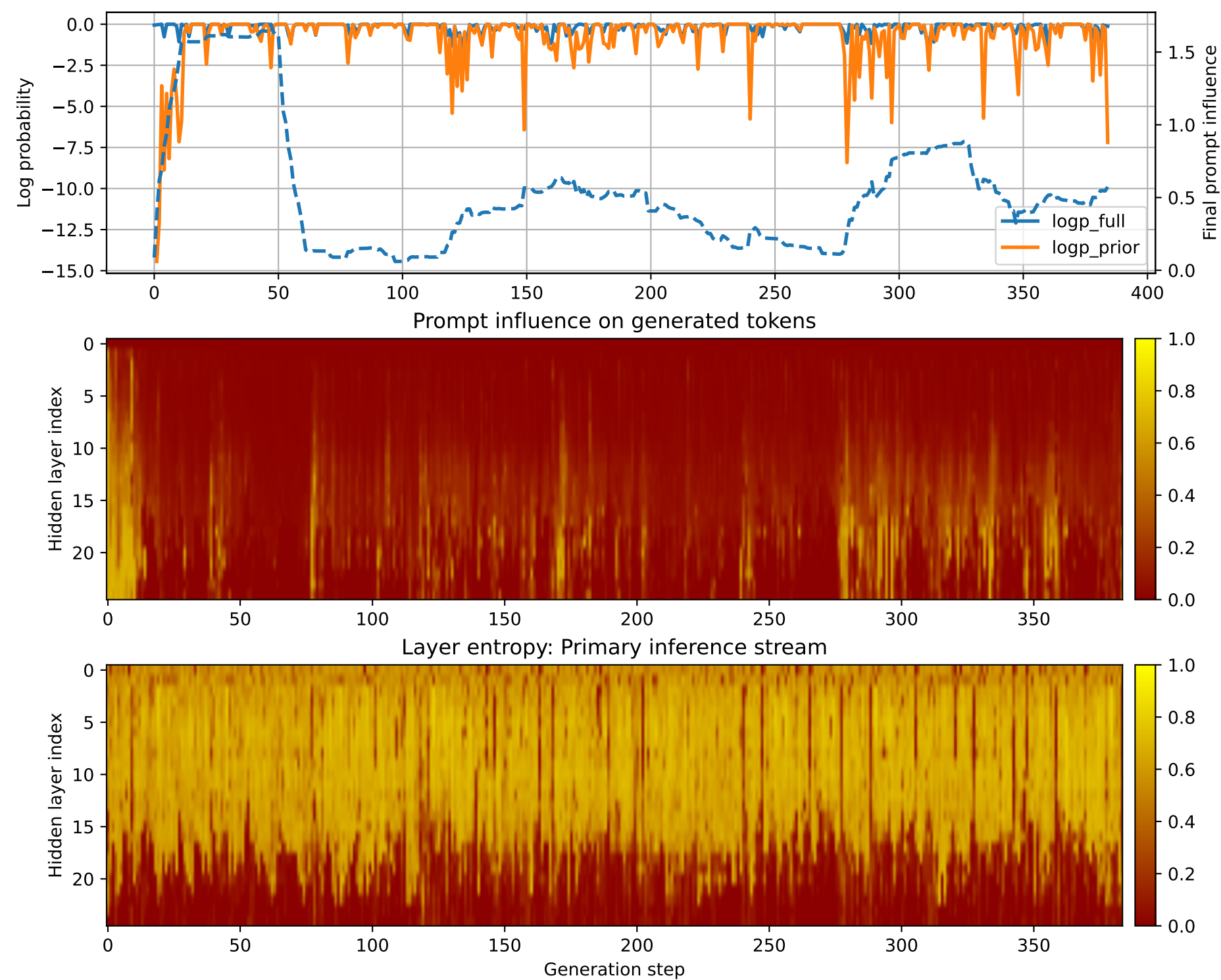


Prompt 1



PROMPT:
Task ID: PYC053 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC053 Your code must contain this required symbol:
class Point Implement this task: define
dataclass Point with distance_to(other).. Start your answer with
the check-token line, then provide the
Python implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC053 class Point: def __init__(self, x:
float, y: float): self.x = x
self.y = y def distance_to(self, other: Point) -> float:
return ((self.x - other.x) ** 2) +
((self.y - other.y) ** 2) # Example usage: point1 = Point(1, 2)
point2 = Point(3, 4)
print(point1.distance_to(point2)) # Output: 5.0 # Explanation:
In the Python code provided, we define a
class called `Point` with two attributes: `x` and `y`. The
`\_\_init\_\_` method initializes these attributes with
the values `x` and `y` respectively. The `distance\_to` method
takes another `Point` object `other` as input
and calculates the Euclidean distance between the two points
using the Pythagorean theorem. The formula for
calculating the distance between two points in a 2D plane is:
distance = sqrt((x2 - x1)² + (y2 - y1)²) Here,
`x1` and `y1` are the coordinates of the first point, and `x2`
and `y2` are the coordinates of the second
point. The method returns the calculated distance as a float
value. In the example usage, we create two
`Point` objects `point1` and `point2` with the respective
coordinates `(1, 2)` and `(3, 4)`. We then call the
`distance\_to` method on `point1` and `point2` and print the
result, which is `5.0`. This indicates that the
distance between `point1` and `point2` is 5 units in the x-y
plane.